Eberhard Karls Universität Tübingen | Seminar für Sprachwissenschaft (SfS)
Methods I: Logic, WS 2026/27 | Dr. Kyle Richardson
Tutors: Aida Rostami, Amit Shekhawat

Exercise Sheet 4
Propositional Logic 2

Exercise 1: [4 points] Translate each of the following natural language sentences into a well-formed formula of propositional logic, using the provided translation key.

(a) 'The professor explained the problem, but the students did not understand it.'

Translation Key:
p – The professor explained the problem.
q – The students understood the problem.
Formula:

(b) 'Neither the professor nor the tutor is available.'

Translation Key:
p – The professor is available.
q – The tutor is available.
Formula:

(c) 'The laboratory is open only if a supervisor is present.'

Translation Key:
p – The laboratory is open.
q – A supervisor is present.
Formula:

(d) 'The students can submit their assignments only if they have registered for the course, and if they submit their assignments, they will receive feedback.'

Translation Key:
p – The students submit their assignments.
q – The students have registered for the course.
r – The students receive feedback.
Formula:

Exercise 2: [4 points] For each formula, indicate which formula in the box is logically equivalent to it by writing the corresponding Roman numeral. Each option is used exactly once.

I	$((p \to q) \wedge (q \to p))$
II	$((p \wedge q) \vee (p \wedge r))$
III	$\neg(p \vee q)$
IV	$(\neg p \vee q)$

(a) $(p \to q)$

(a) _____

(b) $(\neg p \wedge \neg q)$

(b) _____

(c) $(p \wedge (q \vee r))$

(c) _____

(d) $(p \leftrightarrow q)$

(d) _____

Exercise 3: [4 points] For each of the following formulas, determine whether it is a tautology, a contradiction, or a contingency.

(a) $(p \vee \neg p)$

(a) _____

(b) $((p \to q) \wedge (p \wedge \neg q))$

(b) _____

(c) $((p \wedge q) \to r)$

(c) _____

(d) $((p \to q) \leftrightarrow (\neg q \to \neg p))$

(d) _____

Exercise 4: [3 points] Consider the following situations, which represent all possible truth-value assignments for p and q.

Situation	p	q
s_1	1	1
s_2	1	0
s_3	0	1
s_4	0	0

Let $S = \{s_1, s_2, s_3, s_4\}$. Recall that $[\![\varphi]\!]$ denotes the set of situations in which the formula φ is true. Determine each of the following sets.

(a) $[\![(p \wedge q)]\!]$

(b) $[\![(p \rightarrow q)]\!]$

(c) $[\![\neg p]\!]$

Exercise 5: [4 points] For each of the following, determine whether the logical entailment holds. If it does not hold, give a counterexample by assigning truth values to the propositional letters. If it holds, briefly explain why the conclusion follows from the premise(s).

(a) $(p \wedge q) \vDash p$

(a) _____

(b) $(p \vee q) \vDash p$

(b) _____

(c) $(p \rightarrow q),\ \neg q \vDash \neg p$

(c) _____

(d) $(p \rightarrow q),\ (q \rightarrow r) \vDash (p \rightarrow r)$

(d) _____